

ESP32 Dev Module (CH340C)

Complete Library List for Arduino IDE & MicroPython

ARDUINO IDE – REQUIRED LIBRARIES

- 1 esp32 by Espressif Systems (Board Package)
- 2 Wire.h – I2C communication
- 3 SPI.h – SPI support
- 4 Ticker.h – Timers
- 5 EEPROM.h – Non-volatile storage
- 6 OneWire.h – DS18B20 communication
- 7 DallasTemperature.h – DS18B20 driver
- 8 NewPing.h – Ultrasonic sensor
- 9 Adafruit_GFX.h – Graphics core
- 10 Adafruit_SSD1306.h – OLED display
- 11 Adafruit_TCS34725.h – Color sensor
- 12 Adafruit_Sensor.h – Adafruit base sensor
- 13 ESP32Servo.h – Servo motor control

MICROPYTHON – REQUIRED MODULES

- 1 machine (Pin, ADC, PWM, I2C)
- 2 time – delays & timing
- 3 ssd1306.py – OLED display
- 4 onewire.py – 1-Wire protocol
- 5 ds18x20.py – DS18B20 sensor
- 6 hcsr04.py – Ultrasonic sensor
- 7 tcs34725.py – Color sensor
- 8 servo.py – Servo motor control

Note: IR sensor (LM358), LDR (LM358), Joystick, TTP223 Touch, DC Motors (L293D), Buzzer and LEDs do not require external libraries and are handled using GPIO, ADC, and PWM.